Deep Research Prompt: Rebuilding the Consciousness Algorithm (CA) as Project Nexus's Credible Founding Document

How to use this: Paste this whole prompt into a frontier reasoning model with real web search / deep research capability. Attach the current CA draft ("Enhanced Consciousness Algorithm: Neurocognitive GAN Architecture"). This is Nexus's oldest and least-corrected source document — written before any of the citation, claim-status, or falsification discipline now used elsewhere in the corpus existed — and it needs a more thorough rebuild than HTLA or HTNLA required.

Mission, and what "novel" has to mean here

CA is the founding document of Project Nexus. That's exactly why it should get the most careful treatment, not the least. The current draft is not one document — it is at least two unreconciled drafts concatenated together (duplicate section headers, duplicate failure-recovery protocols, the same derived quantity rounded two different ways in its two copies), decorated throughout with parenthetical claims of validation — "(Biologically Validated)," "(Research-Aligned)," "(Dopamine Models)" — that cite nothing, anywhere, across the entire document. It also contains at least one formula that does not compute what it claims to (Section on Φ) and at least one that is calculus-shaped but functionally inert (the STDP integral — see below), plus benchmark tables with real-looking error bars and a specific claimed **24.1% annualized return with a 4.2 Sharpe ratio** in a finance application, none of which traces to any real experiment.

The instruction driving this rewrite is to bring back genuinely new or undiscovered material. Read literally, that instruction is dangerous here specifically, because the likely failure mode — an AI asked to produce impressive novelty producing confident invention instead — is the most plausible explanation for how this document ended up the way it did. So "novel" in this prompt means two specific, checkable things, and nothing else:

Real connections between what CA is already gesturing at and literatures that already exist but that CA never cites — several of which are recent (2021–2023) and directly on point, listed below. Finding and correctly applying a real citation CA is missing is genuine research value, even though the citation itself isn't new.
Precisely stated, honestly labeled open research questions — including at least one direction (Section 6 below) that appears to be a genuinely unexplored connection between active inference and Nexus's own existing epistemic-status work. A well-posed open question is a real contribution. An invented answer to it is not.

Nothing in this prompt should produce a new empirical claim, benchmark number, or validation result that was not actually run. If a genuinely new experiment is proposed, it is proposed as a protocol to execute, not

reported as already having been executed.

Ground truth you must not contradict

The 12-term claim-status vocabulary already standard across the corpus: `CANONICAL, IMPLEMENTED, VERIFIED, PARTIALLY IMPLEMENTED, ACTIVE DEVELOPMENT, PROPOSED, DESIGN TARGET, EXPERIMENTAL, RESEARCH HYPOTHESIS, APPLICATION-SPECIFIC, HISTORICAL/SUPERSEDED, INFERRED`. Reuse it; do not invent a new one.
The master monograph's §73 ("Consciousness Claims and Category Errors") distinguishes three levels for any consciousness-adjacent name — computational variable, functional analogue, phenomenological claim — and states that evidence for the first does not establish the third. Every biologically- or consciousness-named term in CA (dopamine, Hebbian, Φ, "binding," "phenomenal") gets this tagging applied explicitly.
§72 ("What Nexus Has Not Demonstrated") already disclaims consciousness, sentience, and subjective experience as achieved results. CA's rewrite must not contradict this by asserting a validated consciousness metric.
HTLA's epistemic refinement lattice (§7.5 of the current HTLA rewrite: `EXACT`/`AMBIGUOUS`/`UNKNOWN`/`IMPOSSIBLE` as a lattice of possibility sets, with a proved monotonicity property) is the standing precedent for what a genuinely new, small, well-verified formal contribution looks like in this corpus. Any new formal work in CA should be held to that same bar: constructed explicitly, checked exhaustively or proved, entered in a claim-status audit.

Structural rebuild, before any formula gets touched

Merge the duplicate drafts into one document. Identify every duplicated section (the architecture header appears twice; the failure-recovery protocol appears twice; the hardware table appears twice with inconsistent rounding — e.g., the RK4 solver throughput is computed correctly both times but reported as 8.55 MFLOPs/s once and 8.56 MFLOPs/s once). Reconcile into a single source of truth before doing anything else.
Write an explicit scope statement. CA currently overlaps, without acknowledgment, the responsibilities of HTCE (thought-cloud dynamics), HIGC (constitutional/admissibility enforcement), and THDE (temporal homeostasis). State plainly what CA owns versus what belongs to those subsystems, consistent with the master monograph's component responsibility matrix.
Verify the one piece of arithmetic that is actually derived, so it's on record as correct rather than merely trusted. The NN_dynamic FLOP count (40→256→128→64→35 at 2 FLOPs/weight = 106,880 FLOPs/evaluation; ×4 RK4 stages = 427,520 FLOPs/step; ×20 steps/s ≈ 8.55 MFLOPs/s) has been independently checked for this prompt and is correct. Keep it, cite the derivation, and use it as the standard the rest of the document's numbers should be held to.

Domain-by-domain grounding — investigate each of these

Dopamine and reward prediction error. CA's D1/D2 sigmoid activation curves and Hebbian modulation are gesturing at one of computational neuroscience's genuinely well-established results: the dopamine-as-temporal-difference-error hypothesis (Schultz, Dayan & Montague, "A Neural Substrate of Prediction and Reward," *Science*, 1997). Investigate whether CA's plasticity rule can be honestly re-derived from an actual actor-critic/TD-learning update rather than presented as an independently "validated" sigmoid with unsourced threshold parameters (θ_D1=0.5, θ_D2=0.3). Where real pharmacology is asserted (e.g., relative D1/D2 receptor affinity), find and cite the actual source rather than asserting "(validated)."

Global Workspace. CA's `C_g(t) = softmax(...)` global-workspace mechanism should be connected to the real, current literature it is quietly reinventing: Baars's original Global Workspace Theory (1988), and — importantly — recent deep-learning implementations that already exist: VanRullen & Kanai, "Deep Learning and the Global Workspace Theory," *Trends in Neurosciences* 44(9), 2021 (arXiv:2012.10390), proposing an explicit "Global Latent Workspace" architecture; and Goyal, Didolkar, Lamb, Badola, Ke, Rahaman, Binas, Blundell, Mozer & Bengio, "Coordination among neural modules through a shared global workspace" (arXiv:2103.01197), a working shared-workspace mechanism in neural networks. Also locate and assess more recent (2023+) formal/dynamical-systems analyses of global-workspace-style mechanisms (search for stability or contraction analyses of shared-workspace architectures) as a possible source of real mathematical grounding for `C_g(t)`'s dynamics, which are currently just a bare softmax with a subtracted inhibition term.

Free energy principle / active inference. CA's variational/ELBO machinery (`log P(𝒞) ≥ ... - λ_v(t)·KL[...]`) is the right shape for Friston's free energy principle (Friston, "The free-energy principle: a unified brain theory?," *Nature Reviews Neuroscience*, 2010) but never cites it. Investigate the current active inference literature for how prediction-error minimization is actually formalized, and whether CA's "consciousness ODE" can be honestly reframed as an instance of it rather than a bespoke, uncited construction.

STDP and the Hebbian update. The current formula — `Hebbian_Update(A,B) = ∫ [D1_act·max(K(τ),0) + D2_act·min(K(τ),0)]·A∘B dτ` — integrates over τ, but `A∘B` does not depend on τ anywhere in the expression, so the integral reduces to a constant scalar times `A∘B` and does not implement spike-timing-dependent plasticity in any functional sense. Locate the real STDP literature (Bi & Poo, "Synaptic Modifications in Cultured Hippocampal Neurons," *J. Neuroscience*, 1998, and modern neuromorphic-STDP surveys, especially any applicable to the Loihi-class hardware CA already names) and either derive a version of this update where the kernel actually operates on a real timing signal between events, or state plainly that the current formula is decorative and remove the STDP framing until it does.

Φ and Integrated Information Theory. `Φ_t = Σ MI(X_i; X_{-i})` is not IIT's Φ — it is closer to Watanabe's total correlation / multi-information, a real but different quantity, since it never involves a minimum-information-partition search. Correct the naming and the definitional claim regardless of what else is decided about Φ. Separately: investigate and report the current state of IIT as a scientific framework, including the 2023 open letter (124 researchers, including Dennett, LeDoux, and Baars) calling IIT unfalsifiable pseudoscience, the published exchange in *Nature Neuroscience* (Tononi et al.'s reply; commentary from Gomez-Marin & Seth), and the 2023 Templeton-funded adversarial collaboration against Global Neuronal Workspace Theory, which returned inconclusive results. Whatever CA ultimately does with Φ, it cannot present a fixed numeric threshold (">0.5") as an agreed, validated pass/fail criterion — that misstates the

field regardless of which side of the dispute is right. The Butlin, Long, Elmoznino, Bengio, Birch, Constant, Deane, Fleming, Frith, Ji et al. paper "Consciousness in Artificial Intelligence: Insights from the Science of Consciousness" (arXiv:2308.08708, 2023) is the right model for how to discuss this responsibly: multiple candidate theories, explicit indicator properties, no claimed certainty either way.

A genuinely open, genuinely unexplored question worth stating as such. Active inference already has a formal notion of "epistemic value" — the term in its objective that rewards uncertainty-reducing (information-seeking) action, as distinct from reward-seeking action. As far as this search should confirm, nobody has combined that notion with a *typed* epistemic-status algebra like the one HTLA already built for Nexus (`EXACT`/`AMBIGUOUS`/`UNKNOWN`/`IMPOSSIBLE`, with a proved monotonicity property under refinement). Investigate whether active inference's epistemic value term could be usefully decomposed using Nexus's own epistemic-status categories — e.g., whether "epistemic value" behaves differently depending on whether the uncertainty being reduced is `AMBIGUOUS` (resolvable in principle) versus `UNKNOWN` (structurally unresolved) versus `IMPOSSIBLE` (a detected contradiction). State this as an open question with a proposed formal approach and a falsification criterion if one can be constructed — not as a solved result.

What must simply be removed, not re-sourced

The finance-domain claim of 24.1% annualized return and 4.2 Sharpe ratio. There is no responsible way to retroactively source a specific trading-performance number for a document like this; a Sharpe ratio that high, sustained, would be an extraordinary claim for any real strategy. Remove it outright.

Every other domain-specific "Post-research integration improvements" table entry (robotics decision latency, healthcare AUC and patient-outcome percentage) unless a real, describable experiment produced it. If none exists, these become `DESIGN TARGET` at best, with the reasoning for the specific number stated, or are removed.

Every bare "(Biologically Validated)" / "(Research)" / "(Research-Aligned)" annotation that does not resolve to an actual citation found during this research pass. An annotation that cannot be resolved to a source is deleted, not softened.

Required output structure

Title and abstract calibrated to what is proven, cited, or explicitly posed as an open question — no "validated," "confirmed," or "biologically proven" language without a citation attached to the specific claim.

Scope statement distinguishing CA's responsibilities from HTCE, HIGC, and THDE.

Relationship to Prior Art, covering all five domains above with real citations.

Corrected formal core: dopamine/plasticity re-derived or honestly re-scoped; global workspace connected to real architecture-level precedent; Φ renamed or reformulated with the IIT controversy stated plainly; STDP either fixed to depend on a real timing signal or removed.

The open-question section (item 6 above), clearly marked `RESEARCH HYPOTHESIS`, with a stated falsification approach.

Claim-status audit table covering every equation, threshold, and number in the document.
No benchmark or validation number appears without either a real citation, a described and reproducible experiment, or an explicit `DESIGN TARGET` / `RESEARCH HYPOTHESIS` tag.

Hard constraints

No claim of biological validation, research backing, or empirical confirmation without a specific, real, checkable source attached at the point of the claim — a general "(Research)" tag is not sufficient anywhere in the rewrite.
No performance, accuracy, or financial-return number appears unless it was actually produced by a describable procedure; invented precision (specific decimals, error bars) is treated as a severe defect regardless of framing.
No formula is retained if working through its own notation shows it does not depend on the variables it claims to — the STDP integral is the known instance; check every other integral and derivative in the document for the same failure.
Every biologically- or consciousness-named term is tagged at one of the three §73 levels; none is presented as if it were simultaneously all three.
The IIT/Φ controversy is presented evenhandedly — this document does not take a side in an active, unresolved scientific dispute, regardless of which framing would make Nexus's own claims sound stronger.